

Ashwin Satish

Boston, MA | (857)-832-0951 | satish.ash@northeastern.edu | [LinkedIn](#) | [Github](#) | [Portfolio](#)

EDUCATION

Northeastern University

Master's of Science, Data Science (GPA: 3.83)

Machine Learning, Algorithms, Data Mining, NLP, MLOps

Boston, MA

Sept 2024 - Dec 2026

Christ University

Bachelor's of Science, Computer Science (GPA: 3.53)

Bangalore, India

June 2020 - May 2023

EXPERIENCE

Gyori Lab for Computational Biomedicine - Machine Learning Intern

Jan 2026 - Present | Boston, MA

- Parsed **60GB PubMed baseline (35M records)** via memory-efficient streaming parser; extracted **278,885 trial-paper links** for knowledge graph ingestion.
- Reduced LLM output tokens by 300%+ via custom anchor & sentence-index extractors, preserving full provenance traceability on every data point.
- Implemented data pipeline enriching the biomedical knowledge graph with new relationships improving existing functions.

Northeastern University - Machine Learning Researcher

Jan 2025 - Present | Boston, MA

- Boosted true positives from **2% to 77%** while **maintaining zero false positives** for early-stage Lyme cell detection via a 7-model CNN ensemble (TensorFlow).
- Built a pipeline for reconstruction & masking of **10,000+ cell images**, saving the biology team weeks of manual work.
- Improved accuracy by **13%** via augmentation & L2/dropout regularization, achieving production-ready performance.

Northeastern University - Teaching Assistant (Foundations of Data Science)

Aug 2025 - Present | Boston, MA

- Mentored **300+ students** in linear algebra & statistics for ML & Data Science via problem-solving sessions & **NumPy** demos.
- Conducted **4+ weekly** office hours clarifying topics like constraint optimization, low-rank approximation, & regression.

CarerAI - Data Science Intern

Jul 2023 - May 2024 | Gurgaon, India

- Prevented **2,000+ unnecessary procedures** for customers by uncovering **500+** surgery avoidance cases through KNN analysis of patient records against historical data.
- Leveraged EHR for a **risk-score**, improving classification accuracy by **26.4%**, increasing confidence in surgery avoidance cases.
- Improved KNN's accuracy by **20%** using DBSCAN clustering to remove record outliers, reducing false positives in surgery recommendations.

SKILLS

Programming | Data: Python, SQL, R, Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenCV

Visualization | Cloud: Git, Tableau, Matplotlib, Seaborn, Plotly, AWS, GCP, Azure, PostgreSQL, Snowflake, Docker

AI | Modeling: Transformers, BERT, LoRA, AutoML, T5, LangChain/LangGraph, Kubeflow, Ollama, PydanticAI, MLFlow

PROJECTS

[AI-Powered Pokémon Battle Strategy Engine](#)

- Built & dockerized hybrid **Multi-Agent System (MAS)** integrating **Llama 3.2** with a **deterministic backend** for strategic analysis, ensuring **100% accuracy on game mechanics**.
- Engineered a RAG pipeline via ChromaDB & Sentence-Transformers, **indexing 20,000+ strategy docs** to ground LLM inference in semantic context.
- Developed a Recommender System ingesting **1M+ aggregated usage stats** via custom ETL pipeline for optimal team synergies.

[PantryPilot \(2nd Place | MLOps Expo - Coursework\)](#)

- Contributed to the fine tuning, evaluation & automated deployment (**GCP**) of **Llama 3.2 3B** for personalized inventory recipes, resulting in **50% reduction in dietary violations** via LoRA adaptation.
- Collaborated on DPO feedback with Teacher model (**Llama 3 70B**) for preference labelling, VLM for OCR & Veo 3.1 for video.